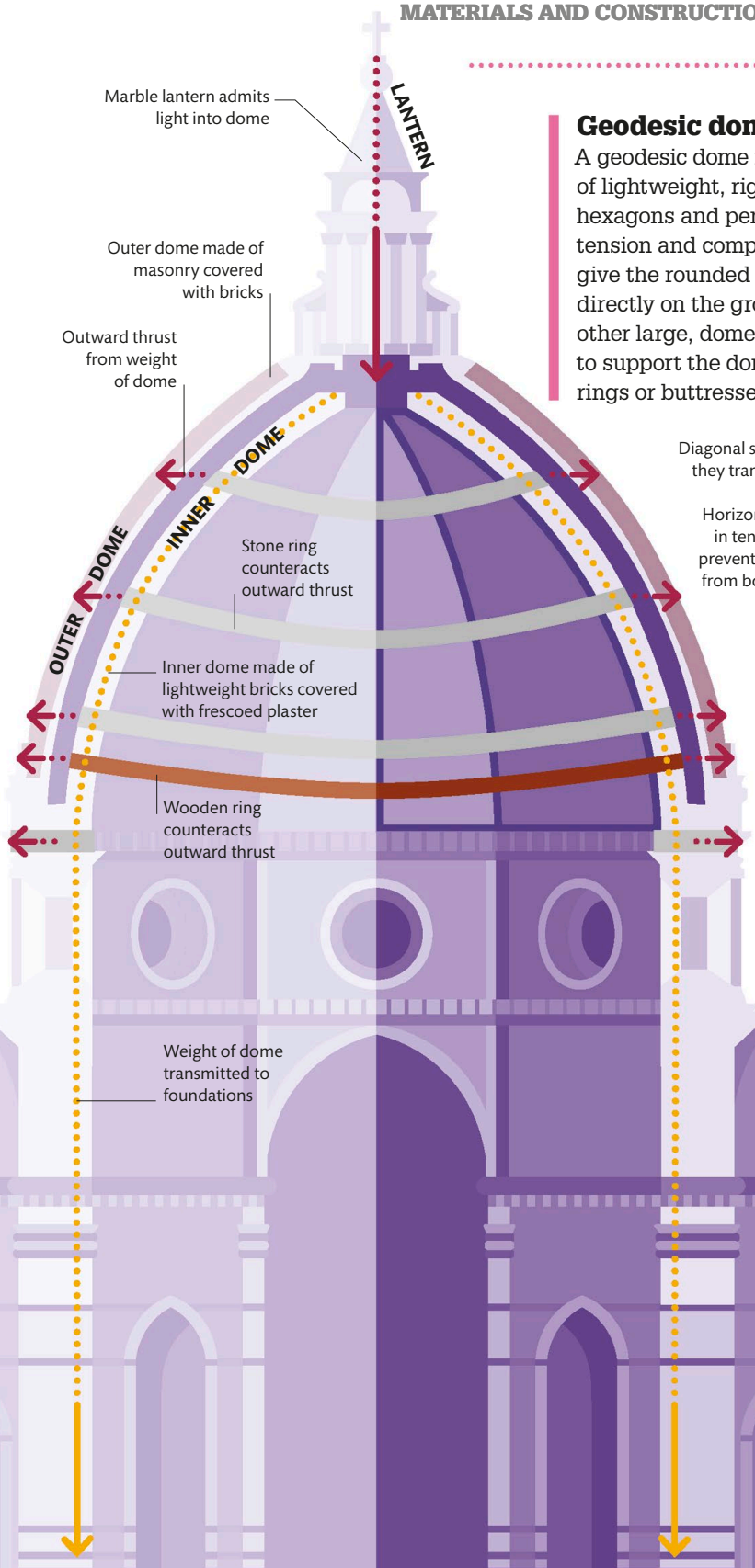
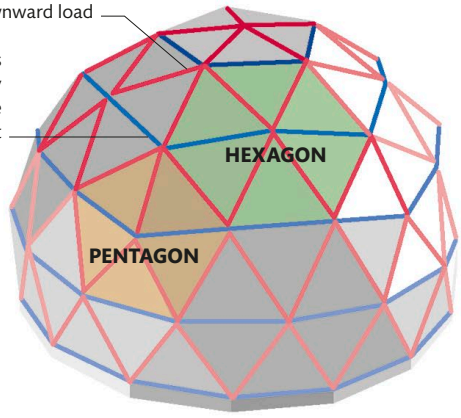
Geodesic domes

A geodesic dome is a spherelike structure constructed of lightweight, rigid struts shaped into triangles within hexagons and pentagons. The triangles are strong under tension and compression. The hexagons and pentagons give the rounded shape. A geodesic dome can be set directly on the ground as a complete structure, unlike other large, domed buildings, which require foundations to support the dome's weight and structures, such as rings or buttresses, to counteract the outward thrust.





Forces in a geodesic dome
The triangular sections act both in compression, carrying the load to the ground, and in tension, counteracting the outward thrust due to the structure's weight.

HOW MUCH DOES A LARGE DOME WEIGH?

A massive dome can weigh several thousand tons. For example, the dome of St. Paul's Cathedral in London weighs about 73,000 tons (66,000 metric tons).